

Full Steam Ahead: Managing Environmental Risk in Canada's Emerging Data Centre Market

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Full Steam Ahead: Managing Environmental Risk in Canada's Emerging Data Centre Market Data centre development is ramping up across Canada, driven by artificial intelligence (AI), cloud computing, and investments in digital infrastructure. What distinguishes data centres from many other forms of industrial development is not just scale, but intensity: facilities operate continuously, draw large and often inflexible electricity loads, rely on water-intensive cooling, and incorporate backup generation and battery systems that introduce emissions and hazardous materials considerations. Given the speed with which data centres are being developed, project proponents and investors are navigating a complex range of environmental issues at the same time that municipalities, utilities and regulators are adapting approval processes to accommodate the development of this relatively new asset class. This article explores the environmental issues and regulatory frameworks that shape project feasibility and approval outcomes, based on recent Canadian experience and lessons from the United States (US), where the sector is more mature. While data centres can raise a broad range of environmental considerations, three issues consistently emerge as the most consequential as projects move from concept to construction: (i) water use, (ii) emissions associated with electricity demand and backup power, and (iii) hazardous materials management. Understanding how and when these issues arise, and addressing them early, can reduce project approval risk and delay. Water Use and Watershed Impacts: The First and Fastest-Rising Pressure Point Cooling is one of the most resource-intensive aspects of data centre operations. Many facilities rely on evaporative cooling systems that consume water by releasing it as vapour, meaning that withdrawals are not returned to the watershed. Even where absolute volumes appear modest, water use can become a focal point for public and regulatory scrutiny, particularly in regions experiencing seasonal water stress. Recent Canadian proposals illustrate this dynamic. In Nanaimo, BC, a proposed data centre drew public concern over potable water use during drought conditions. City staff subsequently concluded that projected water use at full build-out – estimated at between 55,000 and 69,000 litres per day – would be comparable to other commercial and industrial uses in the city. As a result, Nanaimo city staff are working closely with the property owner to ensure the development includes an efficient water cooling system that aligns with municipal conservation objectives. In Alberta, a proposed data centre project in Rocky View County was rejected in part due to concerns about water consumption and stormwater drainage impacts on neighbouring agricultural land. Across jurisdictions, water use is often an early gating issue. Experience from the US shows how quickly

water risk can escalate once multiple facilities operate in the same region. Sector-wide data-centre-related water demand is projected to increase sharply by 2030, with WestWater Research forecasting an approximately 170 percent increase in total U.S. data-centre water demand, driven primarily by cooling needs. At the facility level, some data centres already draw volumes on the order of millions of gallons per day, comparable to the daily water use of small municipalities. Such volumes have prompted public resistance, permitting delays and more stringent approval requirements, particularly in drought-prone jurisdictions. As clusters emerge, regulators and communities increasingly assess water impacts cumulatively rather than on a site-by-site basis.

Emissions from Electricity Demand and Back-up Power: Where Grid Realities Meet Climate Expectations Electricity consumption is central to the environmental profile of data centres. In 2025, Goldman Sachs reported that global data centre power demand is likely to increase by approximately 50 percent by 2027 and potentially by up to 165 percent by 2030. Facilities operate continuously and require a high degree of reliability, which often translates into inflexible electricity loads and extensive backup power infrastructure. Although several Canadian provinces benefit from relatively low-carbon electricity grids, data centres can still increase emissions by driving demand for marginal generation, which is often natural gas-fired. Backup generators, while critical for reliability, also introduce localized air emissions and noise considerations. Canadian experience is beginning to mirror U.S. trends. In Saint John, NB, public concern arose when a proposed data centre was expected to require a substantial share of the output of a nearby natural gas plant, raising questions about greenhouse gas impacts and alignment with regional climate objectives. As electricity demand from data centres has grown, regulators and system operators are increasingly focusing on early engagement, detailed load modelling and the emissions implications of large-load interconnections. In the U.S., where data centre deployment is more advanced, emissions associated with electricity sourcing have attracted significant regulatory and public scrutiny, particularly as AI-driven computing accelerates growth in electricity demand. Canadian utilities and governments are now confronting similar questions around capacity allocation, emissions intensity and long-term system planning.

Hazardous Materials and Emergency Response: An Emerging Area of Focus Modern data centres incorporate materials and systems that carry potentially significant environmental risks, particularly as more facilities are built in the same region. These include large battery systems, fuel storage for backup generators, refrigerants and coolants used in high-density cooling, fire-suppression agents and fluorinated gases in high-voltage electrical equipment. Data centres rely on systems that introduce distinct hazardous materials risks, including battery storage, fuel for backup power, refrigerants and fire-suppression agents. These systems raise fire, spill and contamination concerns, particularly where PFAS-containing materials are involved. In Canada, PFAS are the subject of increasing regulatory scrutiny, including ongoing federal reassessment under the Canadian Environmental Protection Act, 1999 (CEPA), with the federal government signalling an intention to regulate these substances on a class-wide basis. As a result, equipment choices that rely on PFAS-containing materials may warrant closer scrutiny over the life of a data centre project. High-voltage electrical equipment may also contain sulfur hexafluoride (SF₆), a highly persistent greenhouse gas. SF₆ is listed as a toxic substance under Schedule 1 of CEPA, giving the federal government authority to regulate its use and releases. Facilities that exceed greenhouse-gas reporting thresholds must report SF₆ emissions under CEPA's information-gathering provisions. These requirements increasingly influence how equipment is selected, maintained and ultimately decommissioned. As data centres proliferate, the cumulative presence of battery systems, back-up fuels and other hazardous materials are increasingly influencing siting, building design and approvals, as fire authorities, environmental regulators and insurers focus on emergency-response planning, containment measures and long-term liability.

Where Regulation Is Tightening – and Why It Matters for Projects Part 1: Market and Regulatory Signals Shaping Data Centre Development Despite the rapid growth of data centre projects across Canada, no jurisdiction has adopted environmental regulations that are specific to data centres as a distinct project class. Instead, data centres are regulated through existing environmental, energy and permitting frameworks that were not originally

designed with continuous, high-load facilities in mind. As a result, regulatory tensions often arise indirectly, through electricity, water and emissions approvals rather than through data centre-specific rules. Electricity Capacity as the Primary Constraint Electricity capacity has emerged as the most immediate regulatory issue shaping data centre development in Canada. Although no jurisdiction has adopted environmental regulations specific to data centres, regulators have focused on managing the scale and reliability impacts associated with AI-driven facilities. According to the Canadian Climate Institute, unlike traditional data centres, which typically draw between 5 and 10 MW and operate with cyclical demand patterns, AI-driven facilities operate continuously at or near maximum capacity and often require more than 100 MW of electricity (equivalent to the annual electricity consumption of approximately 350,000 electric vehicles). In response, provinces seeking to attract AI and cloud infrastructure have introduced targeted mechanisms to manage large-load electricity demand while protecting grid reliability and advancing broader economic objectives.

Provincial Approaches to Managing Large-Load

British Columbia. Under Bill 31, the Energy Statutes Amendment Act, British Columbia and BC Hydro launched a competitive allocation process in January 2026 to manage electricity demand from AI and data centre development in the province. Prospective data centre projects are required to participate in a competitive selection process administered by BC Hydro. The process is intended to support new development while protecting electricity capacity for established industries such as mining, LNG and forestry, with initial allocation targets for AI and data centre projects of up to 400 MW over the first two years.

Alberta. Alberta has positioned itself as a hub for AI infrastructure while introducing interim limits on large-load connections. The province's AI Data Centres Strategy, released in December 2024, targets up to \$100 billion in private investment over the next five years to develop hyperscale data centre facilities (for more details on Alberta's AI Data Centre Strategy, please see our earlier blog). In June 2025, the Alberta Electric System Operator announced a two-phased connection strategy, including a one-time, interim limit of 1,200 MW for load projects of 75 MW or greater that do not require new transmission system reinforcements or upgrades. A second phase is expected to establish a longer-term framework addressing new reliability standards, tariff redesign, cost allocation reviews and updated forecasting.

Ontario. Ontario has taken a planning-driven approach to data centre growth. As of July 2025, data centres had requested 6,500 MW of new load, representing approximately 30 percent of Ontario's peak demand in 2024 (equivalent to the output of the Bruce Nuclear Generating Station). By 2035, the data centre industry is expected to account for approximately 13 percent of new electricity demand in Ontario. Amendments to the Electricity Act under Bill 40: Protect Ontario by Securing Affordable Energy for Generations Act, 2025, which received Royal Assent on December 11, 2025, will enable the Minister of Energy and Mines to evaluate interconnection applications from specified load facilities, including data centres, based on factors such as economic impact, job creation, and energy intensity. This framework allows connection capacity to be prioritized for projects aligned with provincial objectives.

Québec. Québec has focused on rate reform and continued use of selection processes to manage high-load demand. Peak consumption of large data centres is expected to reach approximately 1,000 MW by 2035. In January 2026, the Québec government directed the Régie de l'énergie to establish new electricity customer categories for data centres and blockchain or cryptocurrency projects. Hydro-Québec has proposed a corresponding rate for large data centres averaging 13 cents per kWh (approximately double what current large power customers pay), with implementation expected in the second half of 2026, subject to approval. Projects exceeding 5 MW remain subject to Québec's existing selection process, and a five-year transitional rate is available to data centres already connected to the grid.

Part 2: Environmental Approvals in Practice

How Environmental Permitting Is Triggered

Environmental permitting requirements for data centres arise under existing federal and provincial statutes and are driven by project-specific characteristics rather than the classification of a facility as a data centre. In practice, permitting pathways are shaped primarily by water use, emissions, and hazardous materials management. Cooling technology, generator capacity, fuel type, load profile and site location determine whether projects trigger environmental assessments,

licences or multi-agency approvals, which may in turn entail additional technical studies, habitat offsetting requirements and consultation obligations. Federal Environmental Approvals At the federal level, approvals may be required where projects affect matters falling within federal jurisdiction such as fish or fish habitat, navigable waters, or federally protected species. Depending on project design and location, this may include approvals under the Fisheries Act, Canadian Navigable Waters Act, Species at Risk Act, or the Migratory Birds Convention Act, 1994. In a data centre context, these requirements most commonly arise in connection with water intake or outfall works, thermal or process discharges, works in or near watercourses, or construction activities that may disturb protected species or migratory birds, which can in turn affect construction scheduling and environmental management planning. Stand-alone data centre facilities do not trigger federal impact assessment. Data centres are not listed activities under the Impact Assessment Act (IAA) or the Physical Activities Regulations, but federal assessment requirements may arise where projects are developed in connection with major transmission infrastructure, large water works or federal lands, or are otherwise designated for review under the IAA. Provincial Environmental Approvals Provincial permitting frameworks operate alongside federal requirements and are generally structured around activity-based approvals for air emissions, water withdrawals and hazardous materials, with environmental assessment obligations triggered only where prescribed thresholds are met or enabling infrastructure is subject to environmental assessment review. In British Columbia, approvals may be required under the Environmental Management Act for air emissions and under the Water Sustainability Act for water withdrawals associated with cooling systems, with permitting requirements assessed on an activity-specific basis. Although stand-alone data centres are not “reviewable projects” under the Environmental Assessment Act, environmental assessments may be triggered where a proposed data centre is coupled with significant enabling infrastructure or falls within prescribed categories under the Reviewable Projects Regulation. In Alberta, approvals under the Environmental Protection and Enhancement Act and Water Act may apply depending on generator capacity, fuel type and water use, with approval requirements tied to specific regulated activities rather than facility type. In water-constrained regions, the availability of water allocations may influence site selection. In Ontario, Environmental Compliance Approvals may be required under the Environmental Protection Act for air, noise and waste-related activities, including standby generation, with eligibility for streamlined registration depending on the nature of the activity and emissions profile. Under the Ontario Water Resources Act, a Permit to Take Water is required where water withdrawals exceed 50,000 litres per day. Data centres with evaporative cooling systems can readily meet or exceed this threshold. Data centre projects are unlikely to trigger an environmental assessment under the provincial Environmental Assessment Act, which generally applies to prescribed infrastructure projects or public-sector undertakings. In Québec, most data centre projects proceed through ministerial authorization under the Environment Quality Act and its implementing regulations, which regulate activities based on environmental impacts. Stand-alone data centres do not trigger the environmental assessment review process unless they are associated with other listed activities. Water withdrawals and certain discharges will require authorization depending on thresholds and classification. Air emissions are governed under the Clean Air Regulation, with standby generation typically representing the most common source of air emissions compliance obligations. Planning Considerations Across the Project Life Cycle Environmental considerations will generally arise at different stages throughout the life cycle of a data centre project, as it moves from site selection to construction and operations. This underscores the value of engaging appropriate environmental expertise throughout the development process. Planning and Site Selection Early decisions largely shape a data centre project’s environmental risk profile. Considerations at the planning and site selection phase include: - Assessing water demand in the context of local watershed conditions and seasonal stress, rather than relying solely on average annual consumption. - Evaluating cooling system options with water use, energy demand and community sensitivity in mind, particularly in water-constrained regions. - Considering electricity sourcing and marginal emissions profiles alongside

grid capacity and the timing of power availability. - Identifying at an early stage whether backup generation or on-site power infrastructure will be required, and how that may affect emissions, permitting and community perception. - Considering cumulative impacts where multiple data centre developments are proposed within the same region, including shared demands on water, electricity and emergency services.

Regulatory Approval Processes As projects move into formal regulatory review processes, environmental assumptions are tested and public scrutiny increases. Strategies to address potential issues include:

- Presenting transparent modelling of water use and emissions, including peak-demand and contingency scenarios rather than only average conditions.
- Integrating stormwater and drainage planning into water use assessments, particularly where agricultural land or downstream users may be affected.
- Providing a comprehensive hazardous materials inventory, covering batteries, fuels, refrigerants, cooling agents and fire-suppression systems.
- Demonstrating containment, spill prevention and emergency response measures that are appropriate to the scale and location of the facility.
- Engaging municipalities, utilities and regulators early in the approvals process to surface potential concerns before formal hearings or consultations.

Construction and Early Operations Regulatory project approvals will typically require proponents to meet certain conditions during construction and operation, including:

- Implementing hazardous materials handling and storage measures as approved, rather than deferring them through value-engineering during construction.
- Ensuring contractors and operators are trained in spill response and emergency procedures, particularly where batteries, fuels or refrigerants are involved.
- Monitoring and maintaining battery systems, generators and electrical equipment in a manner consistent with regulatory requirements.
- Planning for the end-of-life handling and replacement of batteries, refrigerants, PFAS-containing materials and SF₆-containing electrical equipment, including recovery, disposal and recycling considerations.
- Clearly allocating environmental responsibilities among owners, operators and tenants to avoid gaps in oversight as facilities enter service.

Navigating a Complex Regulatory Landscape The development of data centre projects requires proponents to navigate a layered and often jurisdiction-specific regulatory landscape. Complex development issues, including access to electricity and water management, increasingly shape both environmental performance and approval risk. As a result, data centre proponents need to think beyond traditional permitting strategies in order to manage project costs and optimize operations. Our team continues to monitor developments in this fast-evolving space and we regularly advise clients on energy regulatory and environmental considerations relevant to data centre projects across Canada. Please do not hesitate to reach out to our team with any questions on these issues. Stay Connected All form fields are required “*”